ClassOS: Full Topic-by-Topic Plan for Mathematics, Physics and Chemistry (CBSE)

An AI-powered interactive blackboard: what the teacher writes, what the board does, and what the student should understand.

How to read this plan

Every chapter below is broken into topics. For each topic the plan states three things:

- **Teacher writes / draws:** the input the board must recognise.
- **Board response (interactive):** the live demonstration, simulation or solver that opens.
- **Understanding goal:** the concept or common misconception the demonstration is designed to fix. This is the pedagogical reason the feature exists.

Scope note. Chapter lists follow the NCERT/CBSE structure as I know it. CBSE revises its syllabus and NCERT has been issuing new textbooks, so every list must be checked against the current year's CBSE curriculum document before development of that chapter begins. Treat the lists as a content checklist, not a final syllabus. Biology and other subjects follow the same pattern and are left for a later phase.

Principles that apply to every topic

1. **Concept before calculation.** Each topic has a *see it* mode (visual, no numbers), a *build it* mode (students or teacher change values) and a *solve it* mode (worked, step-by-step solution).
2. **Predict, then reveal.** The board can pause and ask the class to guess before the result is shown.
3. **Show the law.** The governing formula is overlaid with live values substituted in.
4. **Misconception cards.** Each topic carries known student errors (for example "heavier objects fall faster") and a demonstration that confronts them.
5. **Verified answers only.** The computer algebra system, simulators and curated databases produce results. The AI explains and never calculates on its own.
6. **Teacher stays in control.** Suggestions appear as small chips beside the writing and nothing opens unless the teacher taps it.
7. **Check for understanding.** Every topic ends with auto-generated quick questions (MCQ, assertion-reason, case-based) built from the board's own state.

PART 1: MATHEMATICS

1.1 Foundation stage, Classes 6 to 8 (compact)

These classes feed everything later, so the board needs at least basic coverage.

Topic	Interactive demonstration	Understanding goal
Whole numbers, integers, number line	Drag a point on a number line, with addition and subtraction shown as moves	Negative numbers as direction and position
Fractions, decimals, ratio, percentage	Drag a slider and see the same quantity as a pie, a bar, a decimal and a percent at once	One quantity, many representations
Playing with numbers, factors, multiples, HCF and LCM	Prime factor tree builds as the teacher writes a number	Why HCF and LCM work
Basic geometry, angles, lines	Protractor and ruler tools, angle pairs highlight as lines are dragged	Angle relationships hold for every position
Triangles, quadrilaterals, symmetry	Drag a vertex; angle sum stays 180 degrees. Mirror and rotation symmetry overlays	Properties are invariant under dragging
Perimeter, area, volume	Cut and rearrange shapes (parallelogram to rectangle, circle to rectangle)	Where area formulas come from
Algebra basics, simple equations	A balance scale: whatever is done to one side is done to the other	Equations as balance
Exponents and powers, squares and roots	Visual squares and cubes growing	Meaning of powers
Data handling, bar graphs, mean, median, mode	Draw or enter data to get graphs and measures	Reading and summarising data
Probability (Class 7 and 8)	Coin, die and spinner experiments with a running frequency graph	Experimental versus theoretical probability

Topic	Interactive demonstration	Understanding goal
Mensuration, factorisation, direct and inverse proportion	Linked tables and graphs showing proportion	Proportion as a constant relationship

1.2 Class 9 Mathematics

Number Systems

Topic	Teacher writes	Board response	Understanding goal
Rational numbers and decimal expansions	`7/13`	Long division animates until the repeating block appears and is marked	Terminating versus non-terminating repeating
Irrational numbers	`sqrt(2)`	Number line zooms endlessly without the decimal repeating; spiral construction of square roots on the line	Irrationals fill the gaps on the number line
Real numbers, operations, laws of exponents	`2^(1/3)` etc.	Evaluates, with exponent laws shown as step rules	Rational exponents
Rationalising denominators	`1/(sqrt(3)+sqrt(2))`	Step-by-step multiplication by the conjugate	Why the conjugate removes the root

Polynomials

Topic	Teacher writes	Board response	Understanding goal
Degree, terms, zeroes	`p(x)=x^2-5x+6`	Graph appears; zeroes marked where the curve crosses the axis	Zero of a polynomial equals x-intercept

Topic	Teacher writes	Board response	Understanding goal
Remainder theorem, factor theorem	`p(x)` divided by `(x-a)`	Long division animates; slider for `a` shows remainder equals `p(a)`	Why the remainder equals the value
Factorisation	`x^3-6x^2+11x-6`	Step-by-step factorisation with the graph confirming roots	Link between factors and graph
Algebraic identities	`(a+b)^2`	Area model: a square split into regions that match each term	Identities as geometry

Coordinate Geometry

Cartesian plane with quadrants. Plot points by writing coordinates or tapping. Reflection and distance from axes are shown by dragging. Goal: coordinates as address.

Linear Equations in Two Variables

Teacher writes `2x+3y=12`. The board draws the line. Tapping any point on the line shows that it satisfies the equation, and tapping off the line shows it does not. Goal: a graph is the set of all solutions.

Introduction to Euclid's Geometry

Axioms and postulates presented as an interactive list. The board shows that Euclid's fifth postulate gives parallel lines. Goal: the idea of axiomatic reasoning.

Lines and Angles

Parallel lines with a draggable transversal. Corresponding, alternate and co-interior angles are colour-coded and the measures update live. A check button confirms the theorem for the current position. Goal: the relationships hold for any position.

Triangles

Congruence criteria (SSS, SAS, ASA, AAS, RHS): the teacher gives three conditions and the board tries to build the triangle. If SSA fails, the board shows two different triangles from the same data. Goal: why some criteria work and SSA does not.

Quadrilaterals

Drag the vertices of a parallelogram while its properties hold. Midpoint theorem shown by dragging a triangle's vertex. Goal: properties as consequences of the definition.

Circles

Drag chords and see equal chords subtend equal angles at the centre. Cyclic quadrilateral opposite angles stay supplementary when a vertex is dragged on the circle. Goal: the theorem holds for every position, not only the drawn one.

Areas of Parallelograms and Triangles

Shear a parallelogram or triangle along a fixed base and watch the area number stay constant. Goal: same base, same parallels, same area.

Heron's Formula

Teacher writes three side lengths. The board computes the semi-perimeter and area, and shows it equals half base times height. Goal: area from sides alone.

Surface Areas and Volumes

Unfold a cone, cylinder or sphere into a net. Fill a cone with liquid to show it takes three fills to fill a cylinder of the same base and height. Goal: where each formula comes from.

Statistics

Teacher enters raw data. The board builds frequency tables, bar graphs, histograms (including unequal class widths) and frequency polygons, and computes mean, median and mode. Goal: choosing the right display and measure.

1.3 Class 10 Mathematics

Real Numbers

Euclid's division lemma animated with blocks. HCF by the division algorithm shown step by step. Fundamental Theorem of Arithmetic shown as prime factor trees. Proof by contradiction for irrationality of `sqrt(2)` presented line by line, with the teacher able to pause at each step. Goal: why a proof forces the conclusion.

Polynomials

Zeroes and the graph: a quadratic with two, one or no real zeroes as the curve is dragged. Sum and product of zeroes verified live against `-b/a` and `c/a`. Division algorithm for polynomials animated. Goal: link of coefficients to zeroes and the graph.

Pair of Linear Equations in Two Variables

Two lines on one graph. The teacher edits the coefficients and the board shows whether the lines intersect (unique solution), coincide (infinite) or are parallel (none), with the condition on `a1/a2`, `b1/b2`, `c1/c2` highlighted. Solves by graphical, substitution, elimination and cross-multiplication, each step shown. Word problems are turned into equations with the variables defined. Goal: algebra and geometry describe the same situation.

Quadratic Equations

A parabola with draggable coefficients. Nature of roots follows the discriminant live. Solves by factorisation, completing the square (visualised with an area model) and the quadratic formula. Word problems (speed, area, numbers). Goal: why the formula works and what the discriminant means.

Arithmetic Progressions

Enter first term and common difference to see terms plotted as dots on a line. The nth term and sum formulas are shown with Gauss's pairing animation. Goal: AP as a linear function over integers.

Triangles (Similarity)

Drag one triangle and a similar triangle scales with equal angles and proportional sides, with ratios displayed. The basic proportionality theorem is shown by dragging a parallel line across a triangle. Areas of similar triangles show the ratio of areas equals the square of the ratio of sides. Pythagoras theorem has an area-rearrangement proof. Goal: similarity as scaling.

Coordinate Geometry

Distance formula as Pythagoras on a grid. Section formula by dragging a point along a segment with the ratio shown. Area of a triangle from coordinates. Goal: algebraic form of geometric ideas.

Introduction to Trigonometry and Applications

A right triangle with a draggable angle: sin, cos and tan ratios update and the board shows the ratio depends only on the angle, not the triangle's size. Unit circle link. Identities are verified numerically for any angle. For heights and distances, the teacher writes the problem and the board draws the angle of elevation or depression scene. Goal: ratios as properties of an angle.

Circles

Tangent at a point shown perpendicular to the radius. Two tangents from an external point are shown equal. Goal: tangent as a limiting case of a secant (animated).

Areas Related to Circles

Sector and segment with draggable angle, with area and arc length changing live. Combination figures (such as a flower bed in a square) shown with the shaded region. Goal: fraction of circle equals fraction of area.

Surface Areas and Volumes

Combinations and conversions of solids: melt a cylinder to form spheres, with volume conserved and shown. Frustum of a cone by cutting a cone. Goal: volume conservation.

Statistics

Mean of grouped data by direct, assumed mean and step-deviation methods side by side. Median and mode with formulas, and cumulative frequency curves (ogives) with the median read off. Goal: why the three methods agree.

Probability

Simulation with thousands of trials for coins, dice and cards, with the running relative frequency approaching the theoretical value. Sample space shown as a grid for two dice. Goal: the law of large numbers.

1.4 Class 11 Mathematics

Sets

Venn diagrams for two and three sets, drag elements in and out, and union, intersection, complement and difference shaded. Cardinality problems with the inclusion-exclusion count.

Relations and Functions

Mapping diagrams with arrows. The board classifies a relation as function, one-one, onto or neither, and shows domain, codomain and range. The vertical line test is shown by sweeping a line over a graph.

Trigonometric Functions

Unit circle with a draggable angle in degrees and radians. Graphs of sin, cos and tan drawn as the point moves around the circle. Sum and difference formulas, double and triple angle, and the general solution of equations shown as repeating points on the circle.

Complex Numbers and Quadratic Equations

Argand plane: complex numbers as points. Addition as vector addition, multiplication as scale plus rotation (modulus and argument shown). Complex roots of a quadratic with negative discriminant. Square roots of complex numbers.

Linear Inequalities

Shaded half-planes for one and two variables, and the intersection region for a system.

Permutations and Combinations

Enumeration visualiser: arrange objects by dragging tiles and see all arrangements generated. Shows why nPr and nCr differ by a factor of `r!`. Grouping and circular arrangement demonstrations.

Binomial Theorem

Pascal's triangle builds row by row, linked to the expansion of `(a+b)^n`. General term and middle term are highlighted.

Sequences and Series

AP, GP and special series plotted. Geometric series partial sums shown converging when the ratio has modulus less than 1. AM-GM inequality shown geometrically.

Straight Lines

Drag slope and intercept. All forms of the equation (slope-intercept, point-slope, two-point, intercept, normal) shown for the same line. Distance of a point from a line, angle between lines, and family of lines through an intersection.

Conic Sections

A double cone cut by a draggable plane gives circle, ellipse, parabola and hyperbola. Each conic is shown with focus, directrix, eccentricity and standard equations. A string-and-pins demonstration for the ellipse. Goal: all four are one family.

Introduction to Three-Dimensional Geometry

Rotatable 3D axes. Plot points, and distance and section formulas in space.

Limits and Derivatives

Limit as an approaching animation from both sides, with a table of values. Derivative as the slope of a secant line becoming a tangent as the interval shrinks. Derivative rules built from first principles. Goal: the limit idea behind calculus.

Statistics

Mean deviation, variance and standard deviation, with data points draggable to show how the spread changes.

Probability

Events as sets on a Venn diagram, with addition rule, mutually exclusive and exhaustive events.

1.5 Class 12 Mathematics

Relations and Functions

Equivalence relation tester (reflexive, symmetric, transitive) on small sets, shown as arrow diagrams. One-one, onto and invertible function checks. Composition of functions with step flow, and binary operations as tables.

Inverse Trigonometric Functions

Graph a function and reflect it across `y=x` to get its inverse. Principal value branches shown highlighted. Goal: why the domain must be restricted.

Matrices

Matrix operations with colour-coded row-by-column multiplication animation. Transpose, symmetric and skew-symmetric. Row operations for the inverse. A transformation viewer shows how a 2x2 matrix moves the plane.

Determinants

The determinant as signed area scale factor in the transformation viewer. Minors, cofactors, adjoint and inverse step by step. Solving linear systems by the matrix method and consistency checks.

Continuity and Differentiability

A graph with a draggable point shows continuity or the type of break (jump, removable, infinite). Differentiability shown as the tangent existing from both sides. Chain rule, implicit, logarithmic, parametric and second-order derivatives, Rolle's and mean value theorems demonstrated geometrically.

Applications of Derivatives

Rate of change problems with linked graphs. Tangent and normal at a draggable point. Increasing and decreasing intervals shaded. Maxima and minima with the first and second derivative test, and optimisation word problems (such as the largest box or smallest fencing) with the shape changing live.

Integrals

Riemann sums with a slider for the number of rectangles and a live area value converging. Indefinite integration methods (substitution, parts, partial fractions, special forms) with verifiable steps. Fundamental theorem of calculus as accumulation animation, and properties of definite integrals.

Applications of Integrals

Area under a curve and between curves shaded, with a parabola, circle or ellipse as the boundary.

Differential Equations

Slope field drawn for a first-order equation with solution curves traced from any tapped point. Order and degree identified. Variable-separable, homogeneous and linear methods solved step by step. Forming equations from families of curves.

Vector Algebra

Rotatable 3D vectors. Addition by triangle and parallelogram laws, scalar projection, dot product as projection, and cross product with right-hand rule and the area of a parallelogram.

Three-Dimensional Geometry

Lines and planes in 3D, with direction cosines and ratios. Angle between lines, shortest distance between skew lines shown as a common perpendicular, and the distance of a point from a plane.

Linear Programming

Constraint lines are drawn from the teacher's inequalities. The feasible region is shaded, corner points are marked and the objective function line sweeps to the optimum. Goal: optimum lies at a corner.

Probability

Conditional probability with a tree diagram that builds as the teacher writes. Bayes' theorem as a reversing-tree demonstration. Random variables with a bar chart of the distribution, mean and variance, and binomial distribution with sliders for `n` and `p`.

1.6 Mathematics engine requirements

- **Exact algebra:** a computer algebra system (SymPy in the browser) for exact arithmetic, solving, calculus and matrices.
- **Step generator:** rule-based templates that follow CBSE answer style, reviewed by CBSE teachers. The AI phrases the steps and never invents them.
- **Dynamic geometry:** a constraint solver so dragged figures keep their relations (JSXGraph or similar).
- **Two-path verification:** every answer is checked symbolically and numerically, and disagreement is flagged instead of shown.
- **Proof support:** line-by-line algebraic validity checking. The board does not attempt full automated theorem proving, and says so.

PART 2: PHYSICS

2.1 How physics works on the board

Physics needs a pipeline that turns a sketch and a few numbers into a simulation.

1. **Scene builder.** The teacher either picks a template (inclined plane, pulley, circuit, lens, spring) or sketches, and writes the values by hand (`m = 5 kg`, `theta = 30 deg`, `mu = 0.2`). The system builds a *scene description* (objects, relations, parameters).

2. **Simulator.** Every simulator reads the same scene description, so recognition can improve without rewriting the simulators.
3. **Overlays.** Force arrows, energy bars, live graphs and the governing formula can be switched on or off.
4. **Assumption banner.** Each simulation states its textbook assumptions on screen (frictionless, ideal string, point mass, no air resistance) and the teacher can switch the assumption off to show the real-world effect.
5. **Units and dimensions.** Every quantity carries units. A dimension checker lets the teacher test a formula, which is a classic CBSE question.

2.2 Classes 6 to 8 Science: physics foundation (compact)

Topic	Interactive demonstration	Understanding goal
Motion and measurement of distances	Ruler and stopwatch tools, distance-time dots	Measuring and describing motion
Light, shadows, reflection	Light source and object with a movable screen, showing shadow size	Light travels in straight lines
Electricity and circuits (basic)	Cell, bulb and switch circuits, open and closed	A complete path is needed
Magnets	Field of a bar magnet with iron filings drawn live	Poles and field
Force and pressure, friction	Push a block over different surfaces and see the effect	Force changes motion
Sound	Vibrating string with frequency and loudness controls	Sound as vibration
Heat and temperature, conduction, convection, radiation	Particle animation of heat flowing	Three ways heat moves
Chemical effects of current, lightning, stars and solar system	Simple animations	Everyday links

2.3 Class 9 Science: physics

Motion

Topic	Teacher writes / draws	Board response	Understanding goal
Distance and displacement	A path with start and end marked	Path length and straight-line displacement are shown, with sign	Displacement is a vector, distance is not
Speed, velocity, average speed	`s = 40 m, t = 8 s`	Computes and animates a moving object, with an unequal-motion option	Instantaneous versus average
Acceleration	`u = 0, a = 2 m/s^2`	v-t graph builds as the object moves	Acceleration as the slope of v-t
Graphs of motion	Teacher draws an x-t or v-t graph	The board animates a matching object, and also converts one graph into the others	Reading graphs as motion stories
Equations of motion	`v = u + at` etc.	Derived graphically from the v-t area, then used in numericals	Where the equations come from
Uniform circular motion	A circle and radius	A moving point with velocity arrow changing direction	Velocity changes even at constant speed

Force and Laws of Motion

Newton's first law: a ball on a track with friction slider from high to zero, showing a body keeps moving without friction. Second law: drag a force and a mass and see acceleration and momentum change, with `F = ma` live. Third law: two carts push apart with equal and opposite arrows. Conservation of momentum in collisions with a before-and-after table. Goal: remove the idea that motion needs a continuing force.

Gravitation

Universal law: two masses with sliders for mass and distance, with `F` changing as inverse square. Free fall: a feather and a coin in air and in vacuum fall together. Weight versus mass on different planets with surface-gravity values. Buoyancy and Archimedes' principle: an object lowered into water, with displaced volume and upthrust measured,

and the sink-or-float decision explained by density. Pressure and thrust shown as force over area. Goal: mass and weight are different, and heavier does not mean faster in a vacuum.

Work and Energy

Work shown as force along the displacement, with the angle slider showing zero work at 90 degrees. Kinetic and potential energy bars on a roller coaster, with total conserved. Power as a rate. Goal: energy changes form but is conserved.

Sound

A vibrating source creates compressions and rarefactions in a particle animation. Frequency, wavelength, amplitude and speed are linked on sliders. Echo and reflection demo with distance. Range of hearing, and ultrasound and sonar applications. Goal: sound is a mechanical wave that needs a medium (silence in a vacuum jar).

2.4 Class 10 Science: physics

Light: Reflection and Refraction

Mirrors: the teacher picks concave or convex and drags the object. The image forms live with ray diagrams, and the board gives position, size and nature of the image. The sign convention is overlaid. The mirror formula and magnification are checked against the drawing. Refraction: Snell's law with a draggable ray and a medium selector, showing the refractive index, the bending toward or away from the normal, and total internal reflection past the critical angle. Lenses: convex and concave with the same live ray diagrams, power of a lens in dioptres. Goal: the three principal rays and the sign convention as a system, not a memorised table.

Human Eye and the Colourful World

Eye model with a lens whose focal length changes. Myopia and hypermetropia are shown with the image falling short or behind the retina, then corrected with a concave or convex lens. Prism dispersion splits white light into the spectrum. Rainbow formation and atmospheric refraction (twinkling of stars, sunrise delay). Scattering explains the blue sky and red sunset. Goal: each phenomenon follows from refraction or scattering.

Electricity

Drag-and-drop circuit builder with cell, resistor, ammeter, voltmeter, rheostat and switch. Charge flow shown as moving dots. Ohm's law with a live V-I graph and a slope that gives resistance. Resistance depends on length, area and material, shown by stretching a wire. Series and parallel combinations with live currents and voltages. Heating effect of current, electric power and energy with a kilowatt-hour bill calculator. Goal: current, voltage and resistance as separate quantities, and series versus parallel as a design choice.

Magnetic Effects of Electric Current

Oersted's experiment shows a compass needle deflecting near a current. Field lines around a straight wire, a loop and a solenoid, with direction by the right-hand thumb rule. Force on a current-carrying conductor with Fleming's left-hand rule. An electric motor animation and the split-ring commutator. Electromagnetic induction with a magnet moving through a coil and a galvanometer, with Fleming's right-hand rule. AC versus DC generator and domestic circuits with fuse and earthing. Goal: electricity and magnetism are linked in both directions.

2.5 Class 11 Physics

Units and Measurement

SI system with unit conversion. Significant figures and error propagation shown on a measurement table. Dimensional analysis checker: the teacher writes a formula and the board checks homogeneity and can derive the relation from dimensions. Goal: units as a consistency tool.

Motion in a Straight Line

Motion graphs linked to a moving object, with area and slope overlays. Relative velocity of two objects, and free fall with air resistance toggle.

Motion in a Plane

Vectors added by drag, with resolution into components on a slider. Projectile motion with launch angle and speed sliders, live path, range, height and time of flight, and a comparison that shows 45 degrees is the maximum range only on level ground. Relative velocity in two dimensions (a boat crossing a river, rain and umbrella). Uniform circular motion with centripetal acceleration vector. Goal: horizontal and vertical motions are independent.

Laws of Motion

Free-body diagrams drawn automatically for blocks on inclines, connected masses, pulleys and lifts. Static and kinetic friction with the angle of repose. Banking of roads. Impulse and conservation of linear momentum. Goal: choose the system and draw every force before writing equations.

Work, Energy and Power

Work-energy theorem with live bars. Conservative forces and potential energy curves with equilibrium points. Spring energy, and elastic and inelastic collisions in one and two dimensions with before-and-after energy. Goal: conservation laws as shortcuts.

System of Particles and Rotational Motion

Centre of mass of draggable particles and of shapes. Torque and angular momentum with a spinning figure. Moment of inertia with the parallel and perpendicular axis theorems. Rolling motion on an incline compared for a ring, disc and sphere (the race). Goal: rotation as the angular counterpart of linear motion.

Gravitation

Kepler's laws with an orbit simulator (equal areas shown as shaded sectors). Gravitational potential and escape speed. Variation of g with height and depth. Satellites: orbital and geostationary, and weightlessness explained.

Mechanical Properties of Solids and Fluids

Stress-strain curve built as a wire is stretched, with elastic limit and fracture. Pascal's law, pressure with depth, Bernoulli's principle with a pipe of changing width, viscosity and terminal velocity of a sphere, and surface tension and capillary rise with a radius slider. Goal: the same law explains many different everyday effects.

Thermal Properties of Matter and Thermodynamics

Temperature scales, thermal expansion, calorimetry with a heating curve showing latent heat plateaus. Heat transfer by conduction, convection and radiation. Zeroth, first and second laws with a piston and a P-V diagram. Isothermal, adiabatic, isobaric and isochoric processes drawn and compared, work as area under the curve. Carnot engine and the efficiency limit. Goal: heat and work are energy in transit.

Kinetic Theory

Particles in a box. Pressure arises from collisions, temperature from average kinetic energy. Gas laws emerge from the simulation. Degrees of freedom and specific heats, mean free path and Maxwell speed distribution.

Oscillations

Spring and pendulum with live x-t, v-t and a-t graphs, energy exchange, damping and driven resonance with a frequency slider. Simple harmonic motion shown as the projection of circular motion. Goal: restoring force proportional to displacement.

Waves

Transverse and longitudinal waves with wavelength, frequency and speed linked. Superposition and standing waves with nodes and antinodes. Vibrating strings and organ pipes with harmonics. Beats and the Doppler effect with a moving source and observer.

2.6 Class 12 Physics

Electric Charges and Fields

Coulomb′s law with two draggable charges. Field lines for any arrangement of charges, with dipole field and torque on a dipole. Gauss′s law with a draggable closed surface showing flux depends only on enclosed charge. Fields of a line, a plane and a sphere.

Electrostatic Potential and Capacitance

Equipotential surfaces overlaid on field lines, and potential energy of a charge system. Conductors in a field with shielding. Capacitors in series and parallel, a dielectric slab inserted into a plate capacitor with changing capacitance and energy shown.

Current Electricity

Drift velocity animation. Ohm′s law at the microscopic level, and resistivity versus temperature. Cells with internal resistance and a terminal-voltage graph. Kirchhoff′s laws on any drawn network with the solver showing currents and voltages. Wheatstone bridge balance, metre bridge and potentiometer experiments as virtual setups.

Moving Charges and Magnetism

Charged particle in a magnetic field traces a circle or helix, with radius and period from sliders. Biot-Savart and Ampere′s law fields for a wire, loop and solenoid. Force between parallel wires. Torque on a loop, the moving coil galvanometer and its conversion to an

ammeter or voltmeter.

Magnetism and Matter

Bar magnet and earth's field, magnetic elements, para-, dia- and ferromagnetism with domain animation, and the hysteresis loop traced as the field is cycled.

Electromagnetic Induction

Faraday and Lenz demos with a magnet and a coil, with the induced current direction shown and the energy conservation argument. Motional emf, eddy currents, self and mutual inductance, and a generator.

Alternating Current

Phasor diagrams with rotating vectors. LCR series circuit with sliders for L, C, R and frequency, showing the resonance curve and the phase angle. Power factor, transformers, and LC oscillations compared to a spring.

Electromagnetic Waves

Oscillating E and B fields propagating at right angles to each other, displacement current, and the electromagnetic spectrum as a zoomable scale with uses.

Ray Optics and Optical Instruments

Mirrors, lenses and refraction at spherical surfaces, with the lens maker's formula. Prisms, and total internal reflection in optical fibres. Microscope and telescope ray diagrams with live magnifying power.

Wave Optics

Huygens' principle with wavefronts propagating. Young's double slit with sliders for slit gap, wavelength and screen distance, and a live fringe pattern and intensity graph. Single slit diffraction. Polarisation with a rotating polariser and Malus's law.

Dual Nature of Radiation and Matter

Photoelectric effect simulator with sliders for frequency, intensity, metal and voltage, and the stopping potential graph built from data. Einstein's equation. de Broglie wavelength with a calculator. Goal: explains why intensity does not change the maximum energy while frequency does.

Atoms and Nuclei

Rutherford scattering with alpha particles hitting a nucleus. Bohr model with an electron jumping between orbits and the emitted line shown in the hydrogen spectrum. Radioactive decay: many nuclei decaying randomly with a smooth exponential, a half-life slider and an activity graph. Binding energy per nucleon curve, fission and fusion.

Semiconductor Electronics

Band diagram of conductors, insulators and semiconductors. p-n junction forming with the depletion layer, forward and reverse bias, the diode I-V curve, rectifier waveforms, Zener diode, transistor as amplifier and switch, and logic gates with truth tables and live circuits.

2.7 Physics engine requirements

- **Mechanics:** a custom RK4 integrator for textbook accuracy, with a rigid-body library used only where collisions and constraints need it.
- **Circuits:** modified nodal analysis, with transient solutions for RC and LR circuits and phasor mathematics for AC.
- **Fields:** grid computation on the GPU for electric, magnetic and potential maps.
- **Optics:** analytic ray tracing for lenses and mirrors, numerical wave computation for interference and diffraction.
- **Thermodynamics and kinetic theory:** particle simulation plus analytic curves, shown together so students see the macroscopic law emerge from microscopic motion.
- **Verification:** every simulation result is checked against its closed-form formula, and any difference above a tolerance is a failed test.
- **Practical work:** all NCERT experiments are provided as virtual procedures with a readings table, graph plotting and error discussion.

PART 3: CHEMISTRY

3.1 How chemistry works on the board

Chemistry has a strict notation, which makes recognition more reliable than in physics, but it needs three views of every substance. The board always lets the teacher switch between them:

- **Macroscopic:** what you see (colour change, gas, precipitate, temperature change).
- **Particle level:** atoms, ions and molecules moving and rearranging.

- **Symbolic:** formulas, equations and graphs.

Students most often fail at moving between these three, so the board links them: tapping a species in an equation highlights it in the animation and in the beaker.

Inputs, in rising difficulty:

1. Formulas and equations (`H2SO4 + NaOH ->`). Subscripts, charges and state symbols are recognised.
2. Compound names ("ethanoic acid"), converted to structures by a name-to-structure tool.
3. Hand-drawn structures. At first use a drawing toolbar (bond tools, atom labels, ring templates), and add recognition of free-hand skeletal drawings later.

Reaction knowledge base. A curated database of every NCERT reaction (reactants, products, conditions, observations, colours, safety note). If a reaction is outside the curriculum, the board says so and does not guess a product.

3.2 Classes 6 to 8 Science: chemistry foundation (compact)

Topic	Interactive demonstration	Understanding goal
Materials and their properties, separation of substances	Mixture beaker with filtration, evaporation and chromatography tools	Choosing a method from the property
Changes around us, physical and chemical	Sort changes into reversible and irreversible with a particle view	What makes a change chemical
Acids, bases and salts (introduction)	Indicator papers dipped in solutions with colour change	Simple classification
Fibres, coal and petroleum, combustion and flame	Flame zones, fuel and oxygen triangle	Conditions for burning
Metals and non-metals	Compare properties, with reactions to oxygen and acid	Properties group elements
Particulate nature of matter	States of matter as particles, heating and cooling	Matter is made of moving particles

3.3 Class 9 Science: chemistry

Matter in Our Surroundings

Particles animation of solid, liquid and gas with temperature slider. Heating curve with melting and boiling plateaus and the latent heat meaning shown at the particle level. Evaporation versus boiling, and diffusion with a dye in water. Goal: particles, not continuous matter.

Is Matter Around Us Pure?

Mixtures, solutions, suspensions and colloids with a light beam showing the Tyndall effect. Concentration as mass percent. Separation techniques (filtration, distillation, fractional distillation, chromatography, crystallisation) as step-by-step virtual setups. Physical and chemical changes, elements, compounds and mixtures sorted by drag and drop.

Atoms and Molecules

Laws of chemical combination demonstrated with mass tables (conservation of mass shown on a balance, constant proportions). Dalton's postulates. Atomic and molecular mass, mole concept as a counting-by-weighing demonstration, formulae from valency with a criss-cross tool and the working shown. Goal: the mole as a counting unit.

Structure of the Atom

Thomson, Rutherford and Bohr models as an animated timeline, with the experiment behind each (alpha scattering). Shells and electron configuration filled by dragging electrons, valency and isotopes and isobars shown for an element. Goal: how experiments forced each model change.

3.4 Class 10 Science: chemistry

Chemical Reactions and Equations

The teacher writes a skeleton equation and the board balances it by solving the atom-count equations, with the hit-and-trial steps shown and an atom-counting table. Reaction types are classified: combination, decomposition, displacement, double displacement, oxidation and reduction. Observations (colour, gas, precipitate) animate in a beaker. Corrosion and rancidity as everyday redox. Goal: conservation of atoms.

Acids, Bases and Salts

Indicators and the pH scale with colours, strong versus weak acid shown as ionisation at the particle level. Reactions of acids with metals, carbonates and bases. Neutralisation with a titration animation. Preparation and uses of common salts (baking soda, washing soda, bleaching powder, plaster of Paris), with water of crystallisation. Goal: hydrogen and hydroxide ions explain acid-base behaviour.

Metals and Non-metals

Physical properties compared. Reactivity series as a ranking board with displacement reactions that the student can predict, then reveal. Ionic bond formation by electron transfer, with a lattice animation. Extraction of metals by activity: roasting, calcination, reduction and electrolytic refining, with the process flow. Corrosion and prevention.

Carbon and Its Compounds

Covalent bonding by electron-sharing animation, with Lewis dot structures built and checked. Allotropes (diamond, graphite, fullerene) in 3D. Versatility of carbon: catenation and tetravalency, with a chain, branch and ring builder. Homologous series, functional groups and IUPAC naming both ways (name to structure, structure to name). Properties and reactions: combustion, oxidation, addition and substitution. Ethanol and ethanoic acid reactions, esterification and saponification, soaps and detergents with micelle cleaning animation.

Periodic Classification of Elements

Dobereiner, Newlands and Mendeleev as a timeline showing why each system was improved. The modern periodic table is interactive, with trend plots across periods and down groups for atomic size, metallic character and valency, and a prediction mode. (If the current syllabus has moved this chapter, the board keeps it as optional content.)

3.5 Class 11 Chemistry

Some Basic Concepts of Chemistry

Significant figures and dimensional analysis. Atomic and molecular mass, percentage composition, empirical and molecular formula by a guided calculator. Stoichiometry with a mole-flow diagram, limiting reagent shown as a ratio of reactant bars, and concentration units (molarity, molality, mole fraction).

Structure of Atom

Discovery of subatomic particles. Bohr model with an electron jumping between orbits and the hydrogen spectrum series. Quantum mechanical model: wave-particle duality, Heisenberg's uncertainty principle, orbitals as 3D probability clouds that can be rotated (s, p, d). Quantum numbers table with the allowed set for any shell. Filling rules (Aufbau, Pauli, Hund) as a drag-and-fill orbital diagram that rejects violations and explains why.

Classification of Elements and Periodicity

Modern periodic law, blocks of the table, and electronic configuration from position. Trends in atomic and ionic radii, ionisation enthalpy, electron gain enthalpy and electronegativity, shown as linked graphs and a heat-map table, with exceptions explained.

Chemical Bonding and Molecular Structure

Octet rule and Lewis structures with formal charge. Ionic and covalent bonding, bond parameters, resonance with structures shown flipping, and VSEPR to predict shapes with 3D rotation. Valence bond theory, hybridisation (sp, sp2, sp3, and d involvement) animated as orbital mixing, and sigma and pi bonds. Molecular orbital theory for diatomics with a filling diagram and bond order. Hydrogen bonding and its effect on properties.

Thermodynamics

System, surroundings and state functions. First law with a piston-and-heat flow animation. Enthalpy of reactions, Hess's law with a cycle builder, and enthalpies of formation, combustion and atomisation. Entropy shown as microstates of particles. Gibbs energy and spontaneity with a temperature slider that flips the sign.

Equilibrium

Dynamic equilibrium as forward and reverse particle rates meeting. Equilibrium constants Kc and Kp with a calculator and reaction quotient comparison. Le Chatelier simulator: change concentration, pressure or temperature and watch the concentration-time curves shift. Ionic equilibrium: strengths of acids and bases, pH with a log-scale demo, common ion effect, buffer action, hydrolysis of salts and solubility product.

Redox Reactions

Oxidation and reduction as electron transfer and oxidation number change. Oxidation-number assignment by step. Balancing by oxidation number and half-reaction methods in acid and basic medium, with step-by-step display. Electrochemical series and redox in titrations.

Organic Chemistry: Some Basic Principles and Techniques

IUPAC nomenclature, structural representations (condensed, bond-line, 3D), isomerism with an isomer generator, electronic displacements (inductive, resonance, hyperconjugation, electromeric) shown with curved arrows, reaction intermediates and types, and purification and qualitative analysis techniques.

Hydrocarbons

Alkanes (conformations with a rotating Newman projection and energy plot), alkenes (geometric isomerism, electrophilic addition with Markovnikov's rule and mechanism animation, ozonolysis), alkynes, and aromatic hydrocarbons (benzene resonance, electrophilic substitution and directing effects).

3.6 Class 12 Chemistry

Solutions

Types of solutions and concentration expressions. Henry's law with a gas-over-liquid simulation. Raoult's law with a vapour-pressure graph for ideal and non-ideal mixtures. Colligative properties (relative lowering of vapour pressure, elevation of boiling point, depression of freezing point, osmotic pressure) shown with a semipermeable membrane animation. Van't Hoff factor and abnormal molar mass.

Electrochemistry

Galvanic cell builder: pick two half-cells and see electrons flow, salt bridge ions move, and the EMF calculated from the standard electrode potentials. Nernst equation with concentration sliders. Conductance, molar conductivity and Kohlrausch's law with graphs. Electrolysis and Faraday's laws, batteries and fuel cells, and corrosion.

Chemical Kinetics

Particle collision animation linking rate to concentration, temperature and catalyst. Rate law and order from experimental data, with a table-to-order solver. Integrated rate laws with concentration-time graphs for zero and first order, half-life, and the Arrhenius equation with an activation energy plot and energy profile diagram.

The d- and f-Block Elements

Electronic configurations and trends across the series, with a periodic table overlay. Properties, oxidation states, and lanthanoid and actinoid contraction. Preparation and reactions of potassium dichromate and potassium permanganate shown as colour-change

animations.

Coordination Compounds

Werner's theory, ligands and coordination number, IUPAC naming of complexes both ways, isomerism in 3D (geometric and optical), bonding with valence bond and crystal field theory (d-orbital splitting diagram that fills with electrons to predict colour and magnetism), and stability and applications.

Haloalkanes and Haloarenes

Nomenclature, preparation and properties. Nucleophilic substitution SN1 and SN2 with curved-arrow mechanism animations, and stereochemistry (inversion and racemisation shown in 3D). Elimination reactions, and the reactivity of haloarenes compared.

Alcohols, Phenols and Ethers

Nomenclature, preparation, physical properties (hydrogen bonding), and reactions with mechanism animations (dehydration, oxidation, esterification). Acidic character of phenols, electrophilic substitution, and ether preparation and cleavage.

Aldehydes, Ketones and Carboxylic Acids

Nucleophilic addition mechanism with the carbonyl carbon electrophile highlighted, named reactions (aldol, Cannizzaro, Clemmensen, Wolff-Kishner, Rosenmund, Etard, Stephen), and qualitative tests (Tollens', Fehling's) as virtual test-tubes. Carboxylic acid acidity and derivatives.

Amines

Classification, nomenclature, basicity ranking with the reason, and preparation (reduction, Gabriel phthalimide, Hofmann bromamide). Diazonium salts and coupling, with a distinguishing test for primary, secondary and tertiary amines (Hinsberg).

Biomolecules

Carbohydrates (glucose open-chain and ring forms with 3D, anomers, glycosidic bond), proteins (amino acid chain builder, primary to quaternary structure, denaturation), enzymes (lock-and-key animation), vitamins, and nucleic acids (DNA double helix with base pairing).

3.7 Cross-chapter chemistry tools

Tool	Used in	What it does
Equation balancer	Class 9 to 12	Solves the atom-count linear system exactly, and shows the CBSE-style method
Mole and stoichiometry calculator	Class 9 to 12	Given masses, moles or volumes, shows limiting reagent, yield and the mole-flow diagram
Molecule viewer	All organic and bonding chapters	3D rotation, bond lengths and angles, orbitals, and measurement
Mechanism player	Class 11 and 12 organic	Step-by-step electron-pushing animation with the teacher able to pause and predict
Named-reaction library	Class 12 organic	Reagents and conditions, with mechanism and an "identify the product" quiz
Virtual lab	All classes	NCERT practicals (titrations, salt analysis, preparation of compounds, chromatography, kinetics experiments) with safety notes and recorded readings
Periodic table explorer	All classes	Trends, configurations, properties and a quiz mode

3.8 Chemistry engine requirements

- **Cheminformatics:** RDKit (WebAssembly) for structures, formulas, masses and substructure matching, plus a name-to-structure converter.
- **Rendering:** a 3D viewer for molecules and orbitals, with 2D depiction for bond-line structures.
- **Balancing and stoichiometry:** exact rational arithmetic, no floating point.
- **Physical chemistry:** shared with the maths and physics engines (rate laws, Nernst, gas laws, thermodynamic cycles).
- **Safety and scope:** the board refuses to predict products outside the curated curriculum database, and demonstrations of hazardous experiments are virtual only.

PART 4: HOW STUDENTS WILL ACTUALLY UNDERSTAND

Having every topic on the board is not the same as students understanding it. These layers are built on top of the topic modules.

4.1 Teaching modes on every topic

Mode	What happens	Why it helps
See it	A clean visual with no numbers, run by the teacher	Builds intuition before symbols
Predict	The board freezes and asks the class to vote on what happens next (via student devices or a show of hands), then reveals	Surfaces misconceptions before they harden
Build it	Teacher or student changes values and watches the effect	Learning by exploring
Derive it	The formula is built step by step from the demonstration (for example the equations of motion from the v-t graph)	Formulas become conclusions, not facts to memorise
Solve it	Worked example in CBSE answer format with marks shown per step	Prepares for board exams
Check it	Auto-generated questions from the exact board state	Immediate feedback

4.2 Misconception library

Each chapter carries a list of known student errors, each paired with a demonstration. Examples: heavier objects fall faster (vacuum drop), a force is needed to keep an object moving (frictionless track), current gets used up in a bulb (ammeters before and after), plants take food from the soil in later modules, dissolving means disappearing (particle view), and a bigger exponent makes a larger number (fractions and negatives). Teachers can add their own entries.

4.3 Question and assessment types

Aligned with the CBSE pattern, including competency-based items:

- MCQ, assertion-reason, and case-study questions built from the board's live scenario
- Short and long answer with a marking-scheme-style model answer
- Numericals with randomised values, so each student gets a different version and the solution is checked automatically
- Diagram-based questions where the student labels or draws on their own device
- Previous-year question bank tagged by chapter, topic and difficulty
- A weak-topic report per student and per class (kept private to the school, in line with India's DPDP Act 2023)

4.4 Language and accessibility

- English and Hindi first, then other regional languages for the interface, explanations and handwriting recognition
- Bilingual notation toggle (for example Hindi numerals)
- Colour-blind-safe palettes and adjustable text size on the large panel
- Read-aloud explanations and captions
- Pen and finger input both work, with palm rejection

4.5 Teacher workflow

- **Lesson planner:** choose a chapter and the board loads a suggested sequence (hook, see it, predict, derive, solve, check).
- **Blackboard mode:** always available. If the teacher wants a plain board, the AI layer is off.
- **Record and replay:** strokes, widgets and voice are captured. The board produces class notes and a replay that absent students can follow.
- **Student devices (later phase):** join by QR code to vote, answer and receive the board's final state.
- **Teacher correction loop:** any wrong recognition is fixed in one tap, and the correction improves future recognition for that teacher.

PART 5: ACCURACY, QUALITY AND ROLLOUT

5.1 Accuracy rules

1. **The CAS, simulators and curated databases are the source of truth.** The AI model explains, and never calculates.
2. **Two-path checks.** Each answer is verified by a second method, and disagreement is shown to the teacher as a warning.
3. **Golden test set.** Every NCERT worked example and exercise question, and recent CBSE papers, become automated tests. A chapter does not ship until the board solves or simulates every item correctly, and chapter content is reviewed by a CBSE-experienced subject teacher.
4. **Out-of-scope honesty.** If the board is not confident, it says so.
5. **Assumptions visible.** Idealisations are always stated on screen.

5.2 Suggested build order

Because chapters are many, release in stages that each deliver a complete useful slice.

Phase	Scope	Notes
1	Shared base: ink, recognition chip, widget framework, variables, units	Everything depends on this
2	Class 9 and 10 Mathematics	Most shared infrastructure, largest teacher demand
3	Class 9 and 10 Chemistry, then Class 9 and 10 Physics	Chemistry grammar is stricter, so reliable sooner
4	Pilot in a few schools	Measure recognition errors and unused features
5	Class 11 and 12 Mathematics	Calculus lab, 3D and vectors
6	Class 11 and 12 Physics	Mechanics first, then electromagnetism and optics, then modern physics
7	Class 11 and 12 Chemistry	Physical chemistry, then organic mechanisms

Phase	Scope	Notes
8	Classes 6 to 8 foundation and other subjects	Extends reach, reuses the same engines
Ongoing	Free-hand diagram and structure recognition, offline models, regional languages	Improves quietly behind stable simulators

5.3 Honest limits

- Recognition of messy handwriting, hand-drawn circuits and hand-drawn organic structures will take longest, so templates and toolbars come first.
- Some topics need very careful pedagogy (for example quantum ideas and organic mechanisms) and will need teacher review before release.
- The curriculum changes, so content must be structured as data (a chapter manifest) that can be updated without code changes.
- The chapter lists in this plan must be validated against the current CBSE syllabus before each phase starts.